

Emerging opportunities for high-temperature solid-state and gas-cycle heat pumps

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Industrial-sector decarbonization requires the adoption of energy-efficient heating technologies such as heat pumps. Among these, vapour compression is the most efficient method. However, its refrigerants pose environmental and safety concerns and preclude heat-pump operation above 600 K. Many industrial processes operating above this temperature use fossil fuels or resistive electrical heating, which generate a substantial amount of unused waste heat. It is therefore essential to develop technologies that efficiently recover and pump heat at such high temperatures. In this Review, we highlight the opportunities and challenges for emerging and environmentally friendly high-temperature heat-pump technologies based on solids or gases. These technologies have the potential to deliver heat at temperatures up to 1,600 K. We provide an outlook on potential solutions, applications and scalability and a roadmap for future technological progress.

Approximately 50% of the world's final energy consumption is used for heating and cooling, with industrial processes accounting for almost half of this energy (Fig. 1a)¹. The majority of heating processes in industry rely on fossil fuels, with only 10% of the energy sources being renewable¹. A large amount of energy is consumed in high-temperature industrial processes. For example, processes that operate below 150 °C account for 7% of the global final energy consumption, processes that operate between 150 and 400 °C account for 5%, and processes that operate at temperatures above 400 °C represent 12%¹.

Moreover, approximately 50% of the global energy consumption results in waste heat (Fig. 1b)². Its distribution by temperature range and relative to the global energy consumption is as follows: 16% is below 100 °C, 4% falls within the 100–300 °C range and 5% is above 300 °C

(ref. 3). Importantly, despite high-temperature waste-heat sources corresponding to a smaller percentage than do low-temperature sources, high-temperature waste-heat sources account for more exergy. Waste-heat source and high-temperature heat-pump utilization are key technologies for decarbonizing industrial heating⁴.

Heat pumps represent an energy-efficient solution for integrating different energy sectors^{5,6} and can be incorporated into Carnot battery systems⁶. They can function as direct heaters, or alternatively as preheaters, recovering heat and increasing temperatures with the aid of an additional heat source and thermal storage systems.

Vapour compression heat pumps, which offer high second-law efficiency and the highest power density per mass of device, currently represent the most significant market-ready technology. However,

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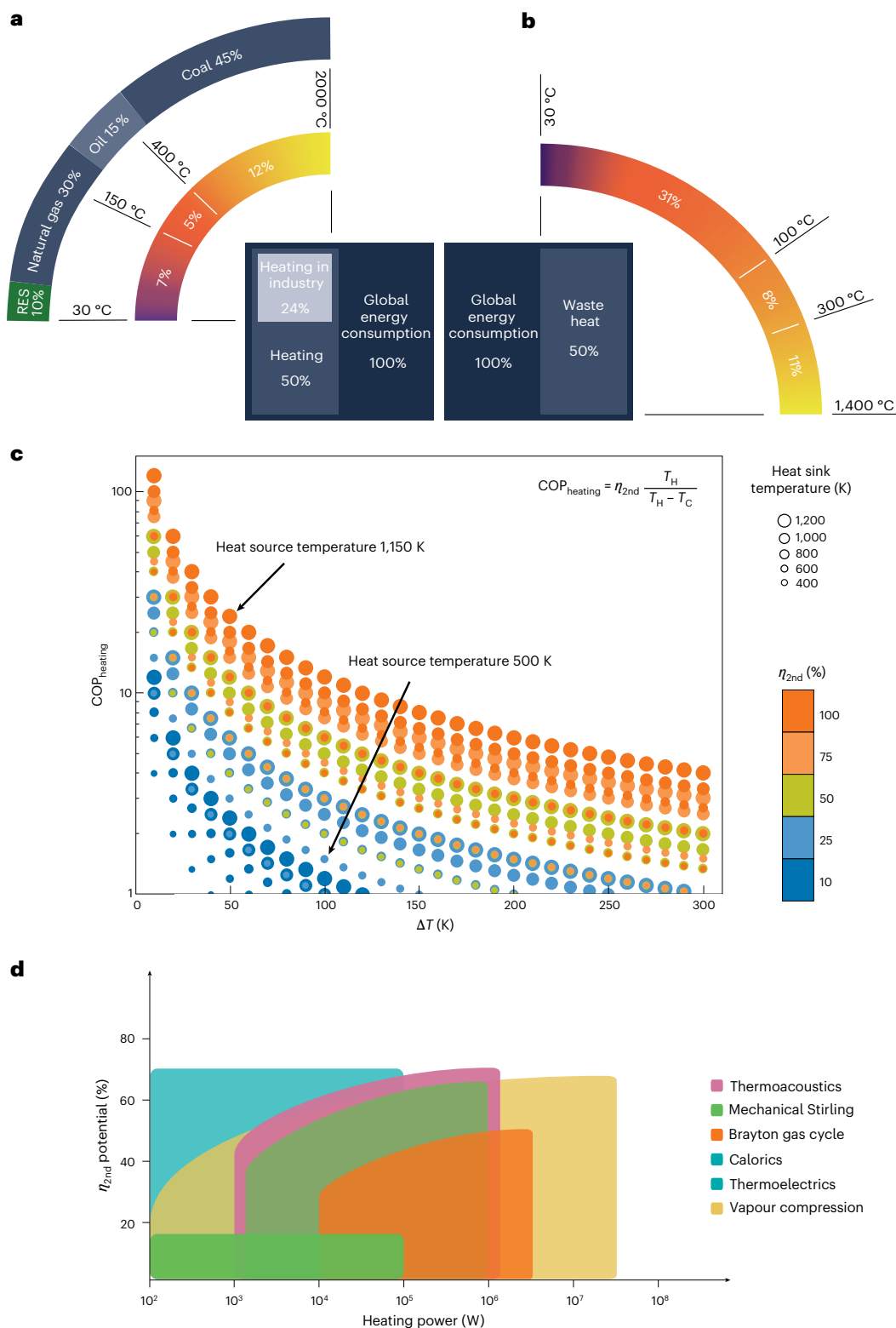


Fig. 1 | Statistical data on global energy consumption and waste heat, and illustrative example of a high-temperature heat-pump energy chain. **a**, Industrial heating share in terms of global energy consumption. The overall share is broken down by temperature ranges corresponding to different industrial processes. The energy source share used for industrial heating is shown, and it comprises renewable energy sources (RES), natural gas, oil and coal. Data are compiled from ref. 1. **b**, Global waste heat share with respect to global energy consumption and the proportion across different temperature ranges. Data are compiled from refs. 2,3. **c**, Performance figures of merit

for different operating parameters of the evaluated technologies: heating coefficient of performance (ratio of the heating power to the power input to the device) COP as a function of the heat sink temperature and temperature difference between the heat source and heat sink for different second-law efficiencies (note: $COP_{\text{heating Carnot}} = T_H / (T_H - T_C)$, $COP_{\text{heating}} = \eta_{2nd} COP_{\text{heating Carnot}}$). **d**, Our view of the future second-law efficiency and heating power for the evaluated technologies for thermoacoustic and for mechanical Stirling^{13,15,18,64,73,78,79,82}, Brayton gas cycle^{16,43,47,48,92}, caloric^{24,28,29,97}, thermoelectric^{20,36,58–61} and vapour compression^{9,98,99} devices.

owing to the physical properties associated with temperature levels, vapour (re)compression heat pumps are currently limited to heating temperatures of up to approximately 550 K (refs. 7,8). They are also limited by refrigerant environmental impact^{9,10}, their flammability or the need for high pressures and associated construction difficulties¹¹. A vapour compression technology alternative, albeit with a significantly smaller market share, is type II absorption heat-pump transformers¹². The difference between a conventional absorption heat pump and a type II absorption heat transformer lies in their operation. Conventional absorption heat pumps utilize the high-temperature heat in the generator to produce a larger amount of useful heat at a medium temperature. In contrast, type II heat transformers upgrade heat from a medium temperature to a higher temperature (exergetic value), but deliver a lower amount of heat compared with the power supplied to the generator.

A gap persists between the industrial demand, available waste heat and other thermal sources, and the technologies required to overcome existing heat-pump temperature limitations. Further advancements are necessary to increase the scalability of low-power applications and improve heat-pump cycle energy efficiency.

In response to these challenges, in this Review, we assess current technologies, challenges, potential solutions, applications, scalability and technology readiness, and present a roadmap for future development of solid-state and gas-cycle high-temperature heat pumps. Some emerging technologies have benefited from decades of research, but their performance near room temperature is still far from that of incumbent technologies. At present, there are no practical alternatives for heating in the range of 600–1,600 K. High temperatures allow for the use of abundant materials with significant thermal effects, so we propose that the exploration of high-temperature heat-pump technologies should advance research and development agendas.

Heat pumping beyond 600 K

Despite their potential, the technologies evaluated in this Review for industrial heating applications above 600 K have been somewhat overlooked. A number of developments for gas-cycle mechanical Stirling¹³ and reverse Brayton-type¹⁴ systems are evident. However, they remain the subject of research activities^{15,16}. A small number of studies have been conducted on high-temperature thermoacoustic gas-cycle heat pumps¹⁷, and only a subset of these studies focused on medium-temperature thermoacoustic heat pumps^{18,19}. Solid-state high-temperature heat pumps represent another alternative, but to the best of our knowledge there are currently no published studies for applications at very high temperatures. For example, whereas numerous studies have focused on thermoelectric energy harvesting²⁰ at high temperatures, studies of Peltier heat pumps have focused on near-ambient-temperature applications²¹. Similarly, emerging solid-state caloric technologies, such as magnetocaloric^{22,23}, electrocaloric^{23,24}, mechanocaloric²⁵ and multicaloric²⁶ heat pumps, have been investigated primarily at low and ambient temperatures.

High-temperature heat pumps are more energy efficient than boilers or electric resistive heaters only when the heating coefficient of performance $\text{COP}_{\text{heating}} > 1$ (Fig. 1c). $\text{COP}_{\text{heating}}$ is defined as the ratio of the heating power to the input power. It can be calculated as the product of the second-law efficiency ($\eta_{2\text{nd}}$) and the ideal Carnot-cycle $\text{COP}_{\text{heating}}$ ($\text{COP}_{\text{heating, Carnot}}$), which is further determined by the ratio of the heat sink temperature (T_{H}) and the temperature difference between the heat source and sink ($T_{\text{H}} - T_{\text{C}}$). For a given temperature difference, $\text{COP}_{\text{heating}}$ increases with increasing heat sink temperature. An increase in the second-law efficiency significantly broadens the operating temperature range with $\text{COP}_{\text{heating}}$ values above 1.

Figure 1d shows our perspectives on the conditions under which the emerging technologies we will evaluate can serve as alternatives to mature vapour compression technology. Notably, the declared efficiencies and power ranges of emerging technologies are the subject

of scientific studies, laboratory experiments or a small number of field tests. Vapour (re)compression heat pumps operating up to 600 K are projected to deliver power from a few kilowatts to tens of megawatts, with second-law efficiencies of 40% for smaller devices and slightly greater than 65% for large devices.

Figure 2 illustrates the different evaluated technologies, their most common thermodynamic cycles and the relationship between the perturbation and response of their refrigerants. The Brayton types of caloric cycle in Fig. 2a–c are discontinuous, as heat absorption/rejection in a single caloric regenerator is not continuous. They consist of the following steps: 1–2 are an isentropic increase in the field; 2–3 are isofield heating; 3–4 are isentropic field release; 1–4 are heat absorption. Increasing the frequency of cycles or the use of tandem regenerators can lead to quasicontinuous operation. The thermoelectric Peltier cycle (Fig. 2d) provides continuous carrier flow through p and n semiconductors, pumping heat from the source to the sink, featuring two quasi-isothermal and two irreversible processes. The Stirling gas cycles in Fig. 2e,g operate discontinuously. At high frequencies, which represent the target operating conditions, they can be regarded as quasicontinuous. The processes labelled 1–2–3–4 include isothermal compression and heating, isochoric regeneration at low volume, isothermal expansion with heat absorption and isochoric regeneration at high volume. The Brayton continuous gas cycle in Fig. 2f consists of polytropic compression, high-pressure isobaric heating and recuperation, polytropic expansion and low-pressure isobaric heat absorption and recuperation.

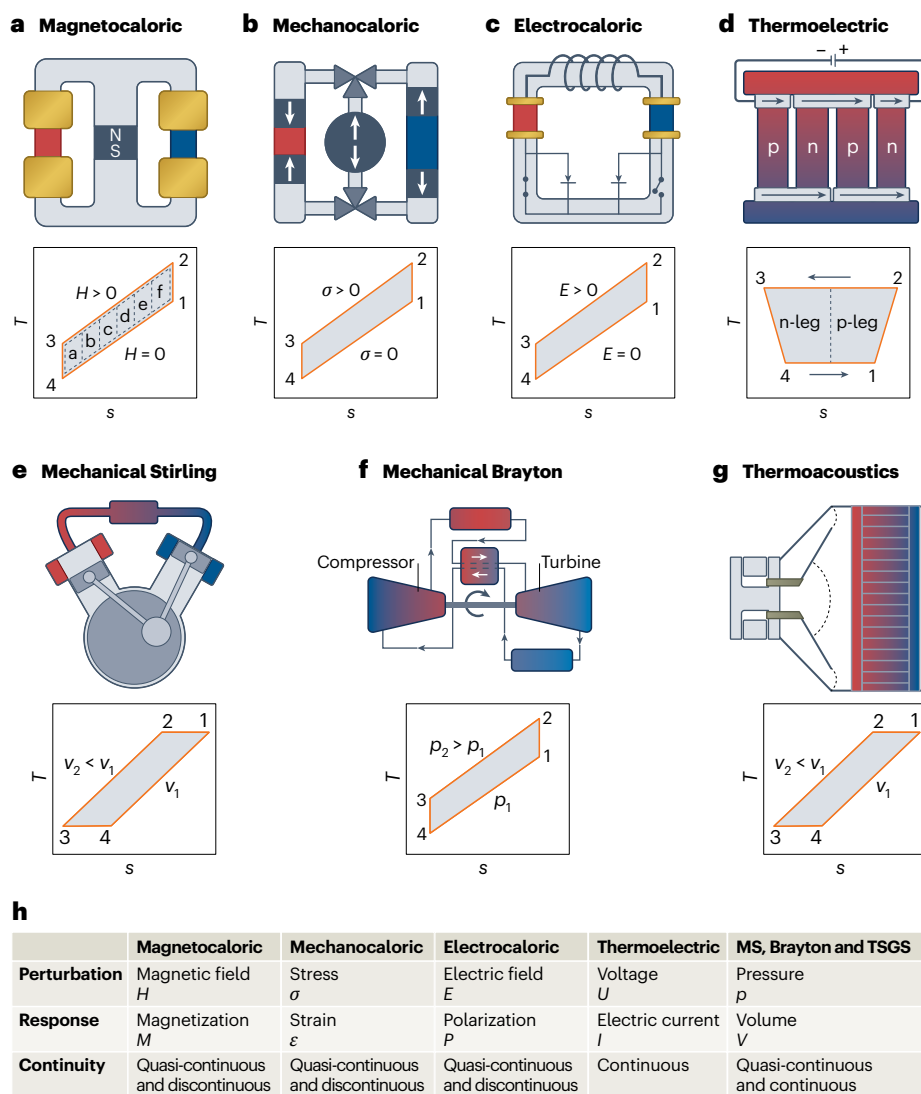
High-temperature solid-state heat pumping

High-temperature solid-state heat pumping offers advantages because of its use of solid refrigerants, which eliminate leakage risks and have recycling potential. Technologies such as magnetocaloric, electrocaloric and thermoelectric (Peltier) heat pumps can operate without moving parts.

Caloric heat pumps

No studies have explored the potential of caloric heat pumps for high-temperature applications. Most prototype caloric device development has traditionally focused on cooling or near-room-temperature heat pumping^{27,28}. The detailed design features and operational characteristics can be found in the following references across related domains: magnetocalorics, refs. 22,23,29; electrocalorics, refs. 23,24; mechanocalorics, which includes barocalorics³⁰ and elastocalorics³¹, ref. 25. In the multicalorics field, which involves the application of multiple driving fields to multiferroic materials, there are currently no prototype devices, aside from a few experimental attempts or simulations²⁶. The solid refrigerant is the most critical component, as it ultimately affects the system performance. The primary caloric material figures of merit are the adiabatic temperature change (ΔT_{ad}) and the isothermal entropy change (ΔS_{T}). For caloric materials to be practical in applications such as heat pumps, minimum values are needed: $\Delta T_{\text{ad}} > 2\text{--}3\text{ K}$ and $\Delta S_{\text{T}} > 5\text{--}10\text{ J kg}^{-1}\text{ K}^{-1}$. The product of the isothermal entropy change and the temperature provides the specific heat absorption capacity, or cooling capacity, which is also a critical characteristic. Conversely, the product of ΔT_{ad} and ΔS_{T} reflects the work absorption capability of the material.

Few magnetic compounds have Curie temperatures of $>600\text{ K}$ (Fig. 3a), with most ferromagnetic materials containing 3d magnetic transition metals (such as Fe, Co, Ni and Mn), and Al as the most common non-magnetic ion. There are a few known high-temperature ferroelectrics (Fig. 3b): lead-based ceramics (lead titanate, lead metaniobate and lead metatantalate) and lead-free ceramics (sodium tantalate, potassium niobate, lithium niobate and Aurivillius-phase ferroelectrics). High-temperature shape-memory alloys, such as Ni–Ti–X (X = Zr, Pt, Hf or Pd)³², are mechanocaloric materials that are useful for high-temperature heat pumps (Fig. 3c), although their



MS, mechanical Stirling; TSGS, thermoacoustic Stirling gas cycle.

Fig. 2 | The most common thermodynamic cycles of the technologies evaluated in this Review. **a–g**, The temperature–specific entropy (T – s) diagrams are denoted by four stages (1–2–3–4). The grey-coloured area inside the T – s diagrams denotes the cyclic work exchanged by the system with its surroundings. In the schematics of the different technologies and their components, the colour scale represents temperature, with red indicating higher values and blue indicating lower values. The designs are made in such a manner that quasicontinuous operation is provided by each of the presented examples. **a**, Magnetic field in the two air gaps of an electro-permanent magnetic field source is provided by coils (gold blocks) near the gaps and a permanent magnet in the central pillar. Red indicates the heated (magnetized) and blue the cooled (demagnetized) state; N and S denote the poles of the permanent magnet. Multiple small-lift materials with different Curie temperatures (**a–f**) may be stacked to achieve a large overall lift, as mostly required in magnetocaloric heat pumps. **b**, Compression (left) induces heating, while extension (right) induces cooling of elastocaloric material. The mechanism provides the forces and the arrows show the force direction. **c**, An applied electric field (left)

induces polarization and heating, while removal of the field (right) induces depolarization and cooling. The change of electric field is controlled with the electronic circuit. **d**, Thermoelectric module consisting of p- and n-type semiconductor legs that are connected electrically in series and thermally in parallel. The bottom surface represents the cold side and the top the hot side. p denotes p-type and n denotes n-type semiconductors. **e**, Gas compression/expansion in alpha-type Stirling device using two pistons to generate heating/cooling, establishing a temperature gradient in the top heat exchanger. **f**, The compressor and turbine share a shaft and change the pressure of the gas inside the device. Gas rejects heat at the hot-side heat exchanger (top) and absorbs heat at the cold-side exchanger (bottom) as pressure changes. An intermediate heat exchanger serves for the heat recuperation of the gas flow between its flow before the expansion in the turbine and before compression in the compressor. **g**, A thermoacoustic loudspeaker (shown in cross-section, left) generates sound waves that compress and expand gas, creating hot and cold regions in the central porous regenerator (right). **h**, The perturbation, response and continuity of the thermodynamic processes for each technology.

caloric properties are very poorly documented. For high-temperature shape-memory alloys, such as Ti–Ni–Pt, Ru–Ta, Ru–Nb and Mn–Pd, one challenge is that repeated reversible actuation is not normally possible due to, for example, creep, phase decomposition and microstructure recovery at temperatures above 770 K (ref. 33). Recent progress has been achieved by developing Ni–Ti–X alloys, with X = Hf, Zr, Pd and Au, that display stable cyclic actuation at 673 K, and Ti–Pd–Cr alloys with very small thermal hysteresis at high temperatures³⁴.

Many mechanocaloric materials that exhibit non-isochoric phase transitions at temperatures of >600 K show good potential for high-temperature applications. Recent examples include $\text{Nd}_2\text{Ti}_2\text{O}_7$ ³⁵, which has a perovskite-like layered structure and displays good resistance to thermal ferroelectric depoling up to 1,673 K. To expand the list of known materials, Fig. 3a–c shows those for which we have found relevant information on the caloric effect in the literature for temperatures exceeding 400 K.

Thermoelectric heat pumps

Multiple devices exist on the market for thermoelectric cooling applications. However, only a limited number of publications exist on prototypes related to thermoelectric heat pumps, all of which focus on low-to-medium-temperature applications²¹. We focused on high-performance thermoelectric materials that present peak performances at medium–high temperatures, that is, above 400 K (Fig. 3d). For this purpose, since the efficiency of thermoelectric devices is correlated with the figure of merit (zT) of the material, we consider only those with $zT > 1$ at high temperatures. A summary of bulk thermoelectric materials can be found in the literature³⁶. For temperatures above 400 K, we can find thermoelectric materials classified into family groups based on Cu–S, Cu–Se or Cu–Te (≥ 800 K, $zT \geq 2$), Ag–SbTe or Ag–Te (≥ 550 K, $zT \approx 2$), GeTe (≥ 700 K, $zT \approx 2$), SnSe (≥ 700 K, $zT \geq 2$), PbTe (≥ 800 K, $zT \geq 2$), skutterudites (≥ 800 K, $zT \approx 1.8$), SnTe (≥ 900 K, $zT \approx 1.8$) or zintlites (≥ 700 K, $zT \approx 1.8$).

High-temperature gas-cycle heat pumping

Thermoacoustic and Stirling heat pumps

High-temperature travelling-wave thermoacoustic heat pumps operating above 200 °C have been envisaged for a decade^{17,37}. Recent numerical investigations predict that these heat pumps can pump heat to temperatures as high as 800 °C (ref. 34). Some prototypes of mechanical Stirling^{13,38} high-temperature heat pumps have been reported to increase temperatures to 550 °C (refs. 18,34,39). For resonant mechanically driven systems, that is, free-piston Stirling heat pumps driven by a linear compressor (for example, a pressure wave generator), double-acting configurations are effective in improving the power density per mass of device and efficiency⁴⁰. For heat-driven thermoacoustic heat pumps, the use of a travelling-wave loop and a multiple-field coordination configuration to optimize the coupling of the energy flow between the engine unit and heat-pump unit is an effective method⁴¹.

A crucial part of thermoacoustic and mechanical Stirling devices is a regenerator, which must provide channels that are significantly smaller than the penetration depth while minimizing viscous losses, ensuring a low axial thermal conductivity and a high heat capacity of solid regenerator materials¹⁸.

For working fluids, low Prandtl numbers and high thermal conductivities are desirable¹⁸. Commonly used gases include helium, nitrogen, argon and CO₂.

Brayton gas-cycle heat pumps

Current large-scale equipment has demonstrated the feasibility of providing temperatures between 300 and 400 °C via reversed Brayton heat pumps, although the number of demonstration systems remains limited⁴². A prototype heat pump consisting of a two-stage axial turbine, a three-stage turbocompressor and three shell-and-tube heat exchangers was reported to provide heat above 250 °C (ref. 43). For pumped heat energy storage with a nominal power of 150 kW, nitrogen was used

as a working fluid in a dual-cylinder reciprocating heat pump/engine, which can reach 750 K in a hot storage tank⁴⁴. Research has focused on optimizing the cycle design⁴⁵, optimizing the heat exchanger to minimize approach temperatures without significant pressure drops⁴⁶ and improving recuperation⁴⁷. The commonly used working fluids are dry air, nitrogen, carbon dioxide or noble gases or their mixtures⁴⁸. They must possess high thermal conductivity, specific volume, heat capacity and pressure ratio. Among the noble gases, argon is notable for its affordability, and helium is notable for its efficiency. Supercritical cycles using CO₂ have been extensively investigated because of their efficiency and compact machinery.

Challenges for solid-state and gas-cycle heat pumps

In Fig. 4 we summarize the strengths and the weaknesses of the various technologies. The primary challenges facing caloric technologies are an insufficient power density per mass of device and a low device efficiency. Among the evaluated technologies, these systems exhibit the least maturity. The heating power density ranges from 10 to 100 W kg_{device}^{−1} for small conventional vapour compression devices and increases to 300 W kg_{device}^{−1} for large-scale implementations²³. Current near-room-temperature magnetocaloric devices achieve heating power densities that barely exceed 1–2 W kg_{device}^{−1} (ref. 49). Much higher values can be found for near-room-temperature electrocaloric devices with electrocaloric refrigerants (>30 W kg_{device}^{−1})⁵⁰. In mechanocalorics, power densities have yet to be reported⁵¹, but promising systems are becoming available^{52,53}.

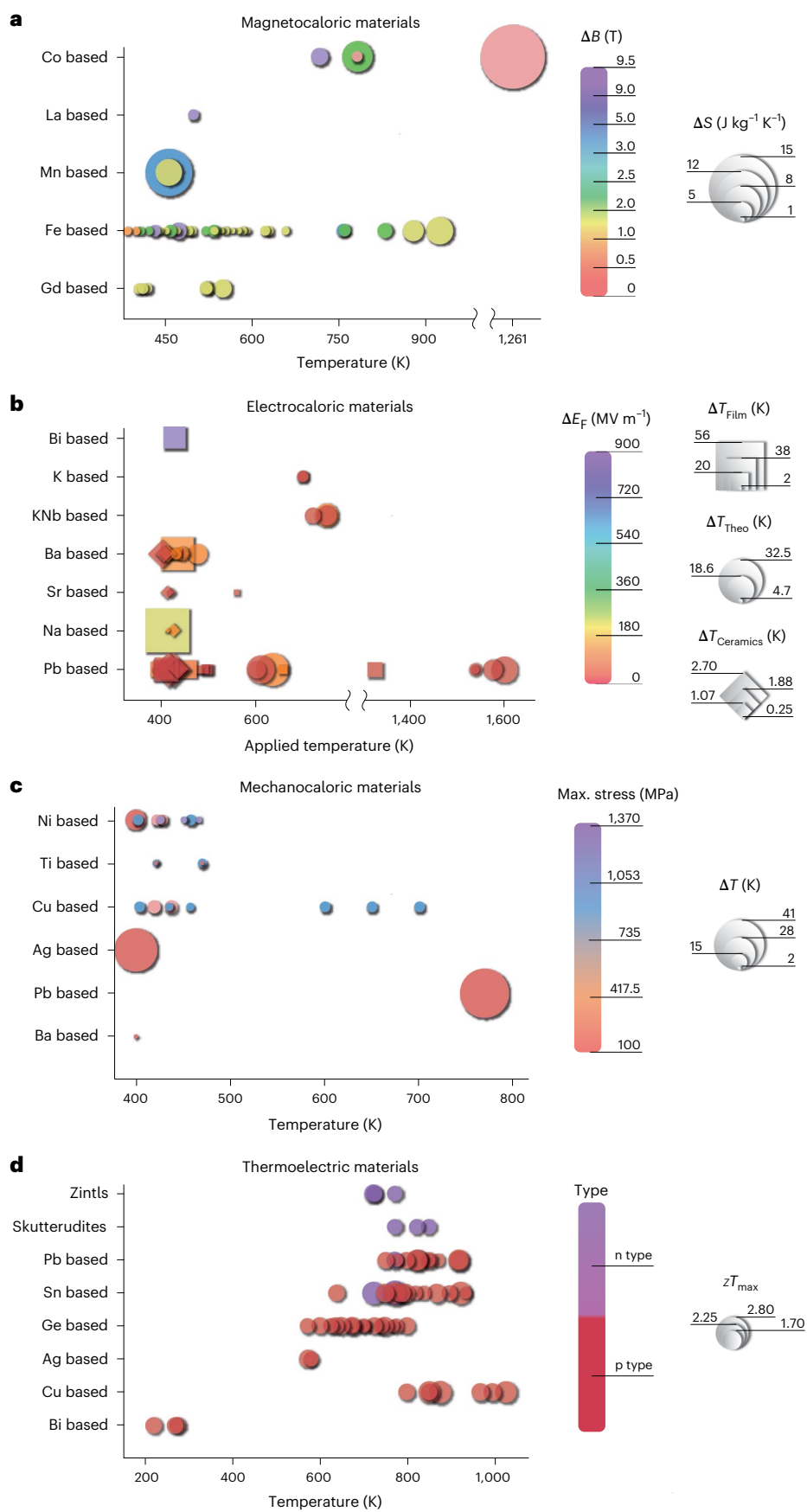
Enhancing the adiabatic temperature change via applied fields, pressure or improved materials is critical but practically constrained. Broadening operational ranges requires optimal material layering, which poses processing and heat transfer challenges. Cyclic stability, hysteresis reduction and high-frequency performance remain problematic. High efficiency demands >80% recovery of field-release energy and field generators (magnetic/electric/stress), which has already been analysed and proved. The maximum experimentally achieved recovered energy release is 91.8% in ref. 54 and 99.7% in ref. 55.

Caloric regenerators require high surface-to-volume ratios with input power viscous losses of <10% (ref. 23). The most used active caloric regeneration principle involves oscillating the fluid through a porous caloric matrix and results in high viscous losses, irreversible heat transfer from rapid cycling, and inefficiencies due to, for example, dead volumes, leakage and valve losses. Therefore, new heat transfer and hydraulic methods are needed. The material abundance, recyclability and carbon footprint must also be considered due to sustainability challenges. Caloric materials can be made by incorporating essentially any ion in the periodic table. The chemical space for designing new caloric materials is vast, and rapid advancements in computational tools promise faster material discovery and optimization in the coming decade.

Fig. 3 | Comparison of the material properties of solid-state heat-pump technologies.

a–d, Magnetocaloric (**a**), electrocaloric (**b**), mechanocaloric (**c**) and thermoelectric (**d**) materials for high-temperature heat pumping above 400 K with properties reported in the literature. These are some representative compositions that show the best performance: **a**, Gd based—Gd₂Fe₁₆Si, GdFe₂Cr₃, GdCo₂; Fe based—Fe₇₅Al₂₅, Fe₇₄Al₂₆, Fe₇₂B₁₂Si₆V₈, Fe₂CoAl; Mn based—MnFeP_{0.67}Ge_{0.34}; La based—(LaFe_{11.4}Si_{1.6})_{5.5}B₁₉Nb₄; Co based—Co₂FeAl, CoFe₂O₄ nanoparticles; **b**, Pb-based ceramics—PMN–30PT (0.7Pb(Mg_{1/3}Nb_{2/3})O₃–0.3PbTiO₃), Pb_{0.89}La_{0.11}(Zr_{0.7}Ti_{0.3})_{0.9725}O₃, 0.7Pb(Mg_{1/3}Nb_{2/3})O₃–0.3PbTiO₃; Pb-based thin and thick films and multilayers—PMN–0.35PT, 0.65Pb(Mg_{1/3}Nb_{2/3})O₃–0.35PbTiO₃, (Pb_{0.88}La_{0.08})(Zr_{0.85}Ti_{0.15})O₃ (1 μm), PZT (lead zirconate titanate); Pb-based bulk—PbTiO₃; Na-based ceramics—(NaBiT–18KBiT); Na-based thin and thick films and multilayers—(Na_{0.5}Bi_{0.5})(Ti_{0.97}W_{0.01}Fe_{0.02})O₃; Sr-based ceramics—SrBi_{1.85}Pr_{0.15}(Nb_{0.5}Ta_{0.5})₂O₉; Sr-based thin and thick films and multilayers—SrBi₂Ta₂O₉; Ba-based ceramics—(Ba_{0.5}Ca_{0.5})(Zr_{0.05}Ti_{0.95})O₃,

0.98BaTiO₃–0.02BiMg_{0.5}Ti_{0.5}O₃, Ba_{0.85}Ca_{0.15}Ti_{0.94}Hf_{0.06}O₃; Ba-based thin and thick films and multilayers—0.5(Ba_{0.8}Ca_{0.2})TiO₃–0.5Bi(Mg_{0.5}Ti_{0.5})O₃; KNb based—KNbO₃; K based—K_{0.5}Na_{0.5}NbO₃; Bi based—0.94(Bi_{0.5}Na_{0.5})TiO₃–0.06BaTiO₃; **c**, Ba based—BaTiO₃; Pb based—PbTiO₃; Ag based—AgI; Cu based—Cu_{59.1}Zn₂₇Al_{13.8}, Cu₂Se, Cu₉Se; Ti based—Ti₇₈Nb₂₂; Ni based—Ni_{42.7}Co_{8.8}Mn_{31.67}Ga_{14.98}In_{2.01}, Ni₅₄Fe₁₉Ga₂₄; **d**, Bi based—Bi_{0.5}Sb_{1.5}Te₃; Cu based—Cu₂Se, Cu_{1.94}Al_{0.02}Se, Cu₂Se + 1 mol% In; Ag based—AgSbTe_{1.85}Se_{0.15}; Ge based—Ge_{0.5}Sb_{0.1}Te, Ge_{0.89}Ti_{0.03}Sb_{0.08}Te, Ge_{0.94}Bi_{0.06}Te + 0.2% nano-SiC, Ge_{0.84}In_{0.01}Pb_{0.1}Sb_{0.05}Te_{0.997}I_{0.003}, Ge_{0.89}Cu_{0.06}Sb_{0.08}Te; Sn based—SnTe + CdTe coating on grains, Sn_{0.81}Ge_{0.05}Mn_{0.2}Te(Cu₂Te)_{0.05}; Pb based—Pb_{0.92}Na_{0.03}Eu_{0.03}Sn_{0.02}Te, PbTe_{0.7}S_{0.3} + 2.5% K, PbTe_{0.8}Se_{0.2} + 8% MgTe; skutterudites—(Sr,Ba,Yb)₃Co₄Sb₁₂ + 9.1 wt% In_{0.4}Co₄Sb₁₂, Ba_{0.3}In_{0.3}Co₄Sb₁₂ + 0.2% Co; zintlites—Mg_{3.15}Mn_{0.05}Sb_{1.5}Bi_{0.49}Te_{0.01}, Mg_{3.02}Y_{0.02}Sb_{1.5}Bi_{0.5}Δ*B*, magnetic field change; Δ*S*, entropy change; Δ*E_F*, electric field change; Δ*T_{film}* and Δ*T_{ceramics}*, the measured adiabatic temperature change of film and bulk ceramics, respectively; and Δ*T_{Theor}*, the theoretically calculated adiabatic temperature change.



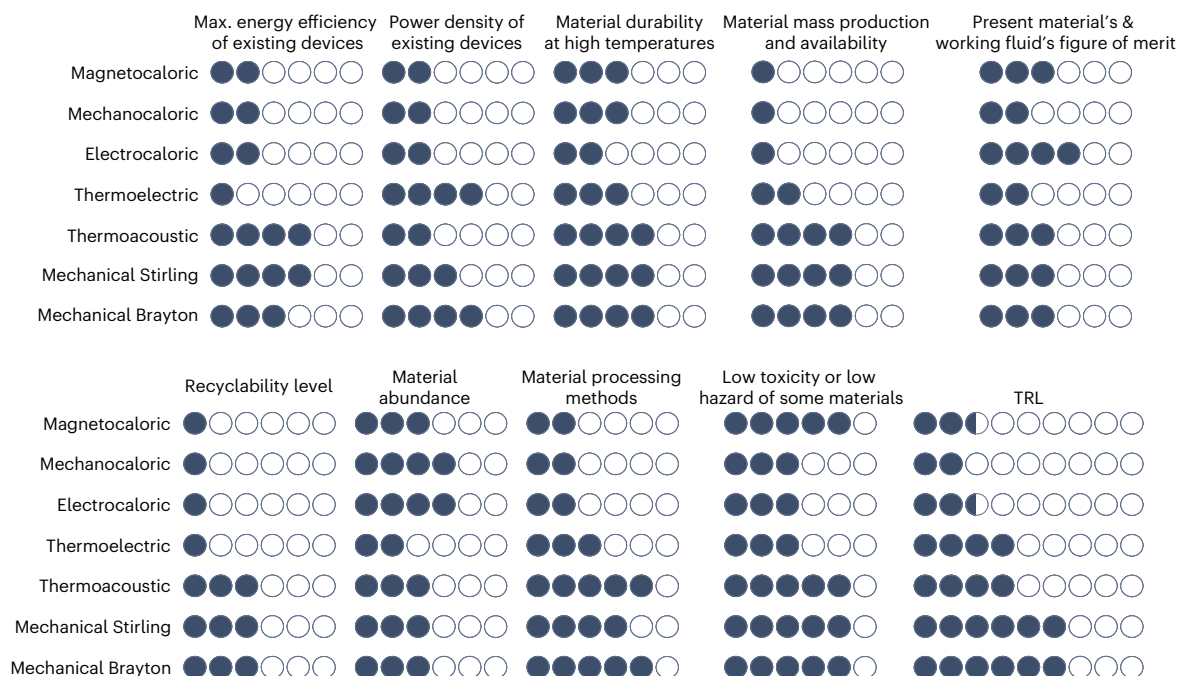


Fig. 4 | Overview and comparison of the evaluated technologies. Indication of necessary improvement activities (white circles) related to the existing development stage (black circles) for the category of the evaluated technology challenges. The last column defines the technology readiness level (TRL). The assessment of the TRL was based on the authors' subjective evaluation of the state of the art of devices, and the maturity of the technology, following the EU definitions and using the available assessment tools¹⁰⁰: TRL 1, basic principles

observed; TRL 2, technology concept formulated; TRL 3, experimental proof of concept; TRL 4, technology validated in a laboratory; TRL 5, technology validated in a relevant environment; TRL 6, technology demonstrated in a relevant environment; TRL 7, system prototype demonstrated in an operational environment; TRL 8, system complete and qualified; TRL 9, actual system proven in an operational environment.

Concerning thermoelectric technology challenges, most efforts have been made to improve the figure of merit of thermoelectric materials, which typically reach values of approximately up to 2 (ref. 56). Market-available Peltier modules offer a high-power density of $>300 \text{ W kg}_{\text{device}}^{-1}$ and substantially exceed this density with thin films⁵⁷, although at a lower efficiency. The large drawback is the small second-law efficiency, in the range of 5–20%, depending on the temperature level^{20,36,58–61}. Scientists focus on efficiency, scalability, stability and longevity, avoiding toxic⁵⁶ or scarce elements. Traditional manufacturing methods include ball milling and printing but must adapt to material types and avoid contamination. Future challenges include thermal and electrical interfaces⁶², novel designs^{36,59} and integration strategies⁶³.

The most important challenges of high-temperature thermoacoustic and mechanical Stirling heat pumps concern materials, processing, manufacturing, working fluids, mechanical components and thermoacoustic parts (especially regenerators). Currently, the power density of thermoacoustic heat pumps is $75 \text{ W kg}_{\text{device}}^{-1}$ (ref. 64). The power density of mechanical Stirling heat pumps is of the order of $60\text{--}100 \text{ W kg}_{\text{device}}^{-1}$ (ref. 65), with an exergy efficiency of approximately 55%, and for devices on a scale between 0.5 and 1 MW (ref. 65). The low power density per mass of device is one of the main challenges for these technologies. Regenerators must be improved to provide a high operating frequency beyond 200 Hz (ref. 66) while maintaining low viscous losses. This is also associated with working fluids, for which abundant fluids with good thermal and viscous properties must be used. Fluids are also related to achieving more efficient heat transfer between the heat source/sink, which is crucial for optimal performance. Another challenge concerns scaling thermoacoustic devices up to a power output of 1 MW and above, including the evenly flowing configurations and components needed for high-power operation, such as heat exchangers and acoustic drivers.

The challenges for high-temperature Brayton gas-cycle heat pumps concern low exergy efficiency (33%) and low power density ($>45 \text{ W kg}_{\text{device}}^{-1}$)⁶⁷. More experimental data on fluid mixtures and the effects of fluid impurities at elevated temperatures, particularly their interactions with materials, heat transfer efficiency and flow conditions, are needed⁶⁸. Challenges in using noble gases such as helium axial compressors include size, stage number, pressure containment and sealing⁶⁹. Another challenge concerns the abundance of helium. Significant efforts must be directed toward high-level recuperative cycle design optimization.

Solutions and future research roadmap

To address the challenges illustrated in Fig. 4 and explore potential solutions for the evaluated technologies, we developed a roadmap outlining the key research activities necessary to achieve future solutions and address upcoming challenges by 2040 (Fig. 5). This roadmap includes a timeline and a description of the main research tasks as well as the most important solutions to existing challenges.

We predict that by 2040, following the roadmap and the solutions outlined in this Review, the importance of the power density per mass of device and $\eta_{2\text{nd}}$ of the evaluated high-temperature heat-pump technologies will increase (Table 1).

The power density per mass of caloric devices can be increased by more than an order of magnitude by increasing the frequency of operation, that is, by implementing thermal switches and thermal diodes^{70,71}, or by high-frequency regenerators^{23,72}. High operating frequencies also demand new types of thermodynamic cycle and novel field source designs. High-frequency regenerators^{23,72} can serve well in thermoacoustic and mechanical Stirling devices and help reach frequencies far above 200 Hz. Scaling up the system via multiunit modular configurations, double-action configurations and multiple stages will also increase the overall capacity and efficiency. Optimization of the

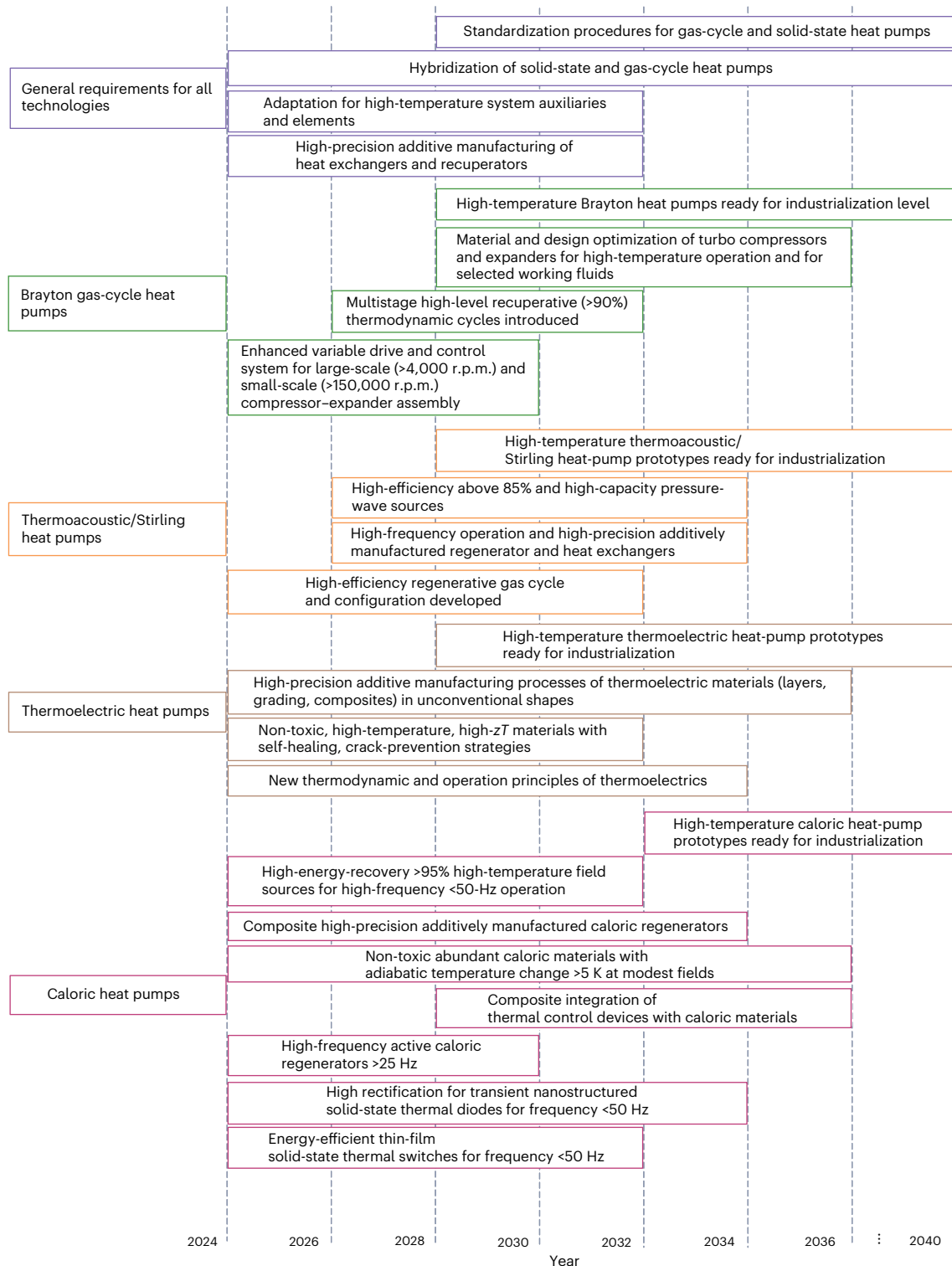


Fig. 5 | Research roadmap for the future development of solid-state and gas-cycle high-temperature heat pumps. The requirements are aligned with the suggested timeline as follows: general requirements for all technologies; Brayton gas-cycle heat pumps; thermoacoustic/Stirling heat pumps; thermoelectric heat pumps; caloric heat pumps.

resonator length⁷³ and an increase in the mean pressure are additional measures for better power density per mass of device²⁹. Similarly, research on Brayton types of heat pump must focus on the elevation of the operational pressures in the system and the frequency of operation. An increase in the power density per mass of the device of the Brayton cycle can be achieved by tailoring the cycle properties to the specific application, including the choice of working fluid, to enable expansion

in the vicinity of the critical point, where the specific volumes and the related expansion work are only a fraction of those of an ideal gas⁷⁴.

To enhance the energy efficiency, the search for and introduction of new materials must represent a continuous process. For all types of evaluated technology, the efficiency can be improved by implementing genetic algorithms and fractal structures of heat exchangers and regenerators. The latter can be further improved by neural

Table 1 | Predictions of power density, efficiency and application power range until 2040 for the evaluated technologies

Technology	Power density per mass of device (W kg ⁻¹)		Second-law efficiency η_{2nd} (%)		Maximum power achieved per device (kW)	Future application power range (kW)
	2025	2040	2025	2040	2025	2040
Magnetocalorics	3	15–20	30	60	15	0.5–50
Mechanocalorics	1.5	15–20	30	60	1.5	1–50
Electrocalorics	30	100–150	55	60	0.01	0.1–10
Thermoelectrics	300	400–500	5–20	20	10	0.1–100
Thermoacoustics	75	200	55	>60	500–1,000	50–1,000
Mechanical Stirling	60–100	300	55	>60	500–1,000	50–1,000
Reverse Brayton	45	150	33	>40	200	100–1,000
Superconducting magnetocaloric (special case)	4	100	50	<70	0.6	50–1,000

Thermoacoustic and mechanical Stirling^{13,15,16,64,73,78,79,82}, Brayton gas cycle^{6,43,47,48,92}, caloric^{24,28,29,51,93,94}, thermoelectric^{20,36,58–61} and superconducting magnetocaloric (special case)^{95,96} devices. The data are predicted for an operating temperature span exceeding 20 K between the heat source and heat sink. The data for year 2025 for mechanocalorics do not include barocalorics due to missing information.

network optimization and further advancements in processing with additive manufacturing.

In calorics, the introduction of static thermal control devices will improve efficiency⁷¹ and will further open the opportunity for hybridization of high-frequency caloric and thermoelectric principles, taking advantage of each. New types of high-frequency/high-field source can further improve efficiency⁵⁴.

In thermoelectrics, different and interrelated solutions should consider holistic approaches in new thermal and electrical interface design⁶², thermoelectric device designs^{59,75} based on optimization of the number of pillars, layered grading and shapes, substrates and/or separators, cyclability and stability, flexibility or miniaturization, among other methods^{36,61,76}. Strategies such as the development of self-healing thermoelectric materials and crack prevention strategies would be highly beneficial in future designs. Pulsed or transient operation⁷⁷ combined with thermal control devices could also contribute to the development of novel strategies that improve the performance of thermoelectric heat pumps^{70,71}.

An improvement in the energy efficiency of thermoacoustic/mechanical Stirling systems can be achieved by additively manufacturing ordered structures of length-optimized⁷³ regenerators. Higher pressures are required for efficiency improvement. Reducing the dead volume in mechanically driven Stirling systems remains an ongoing research task^{73,78}. Replacing solid pistons with liquid pistons can simplify production and reduce viscous losses⁸⁴. Novel mechanisms, including mass-transfer-enhanced thermoacoustic conversion, acoustic radiation and heat pumping in hypersonic boundary layers, should be further evaluated^{79–81}. Researchers should further evaluate improvements in thermodynamic cycles and configurations⁸². Special emphasis must be placed on improving the energy efficiency of pressure-wave sources (pressure-wave generators, piston assemblies) with high levels of energy recovery⁸³ and exploring solutions to complex non-linear acoustic or thermoacoustic effects, including time-averaged secondary steady mass flow (for example, mass-flow streaming) and shock waves⁸⁴. The use of high-temperature ionic liquids or even liquid-metal alloys will improve heat transfer with an external heat source–sink system.

For Brayton-type heat pumps, better efficiency can be reached by implementing levitating axles and improved types of motor, the use of lightweight materials, improved efficiency of the axial compressor and the turbine, and enhanced cooling methods for the rotational components⁸⁵. Further improved heat-exchanger designs with bonded plate fins and 3D-printed structures with integrated ceramic materials and nature-inspired designs are needed.

Operation at high temperatures requires special attention not only for available material resistance but also for functional materials. For caloric technologies, large efforts need to be invested in new refrigerants since the high-temperature domain has been overlooked. Moreover, in the case of magnetocalorics, high-temperature permanent magnets (that is, SmCo) should replace NdFeB magnets. For superconducting magnets, better adiabatic barriers need to be developed between the refrigerant operating at high temperatures and the high-temperature superconductor coil. In electrocalorics, further investigations must reduce the electrical resistance of electrodes at high temperatures. Special attention must be given to composites of caloric materials and thermal control devices and their bonds with the heat-source and heat-sink heat exchangers, which include measures related to thermal expansion.

In thermoelectrics, thermal interface materials should be avoided by direct bonding of thermoelectrics with the heat-source and heat-sink heat exchangers while providing measures for different thermal expansion coefficients of materials. Thermoacoustic/mechanical Stirling heat pumps require the application of high-temperature alloys suitable for high-pressure and high-temperature heat exchangers, and vessels must be researched and developed to ensure durability and performance under extreme conditions. Moreover, the high-temperature permanent-magnet assemblies used in pressure-wave generators⁸⁶ and thermal-buffer-type pressure-wave generators should be utilized to ensure an efficient acoustic wave supply. High-temperature Brayton gas-cycle heat pumps require adaptation of technologies currently employed in the production of high-pressure gas turbines, such as single-crystal superalloys, precision casting techniques and protective coatings. Furthermore, experimental evaluation of materials, especially heat-exchanger ceramic materials, under high-temperature conditions is crucial. Materials must resist corrosion, fatigue and creep under high temperatures and pressures⁴⁶. Oil-free bearings and turbo-compressors require design innovations to extend their operational efficiency and ensure reliable, vibration-free performance⁸⁷.

Sustainable working fluids must be environmentally friendly, abundant, non-toxic and non-reactive and should possess excellent thermohydraulic properties. Thermal oils can provide applications up to 350 °C and can be used as heat-transfer fluids in gas-cycle devices, as well as working fluids in solid-state technologies. They also possess dielectric properties. However, their low thermal conductivity and high viscosity affect their performance, which requires further investigation of high-temperature and high-thermal-conductivity oils. Among gases, the scarcity of helium and molecule-size-related containment issues offset the gains of excellent temperature lift and thermal properties when considering it for gas-cycle or solid-state heat pumps. As an alternative,

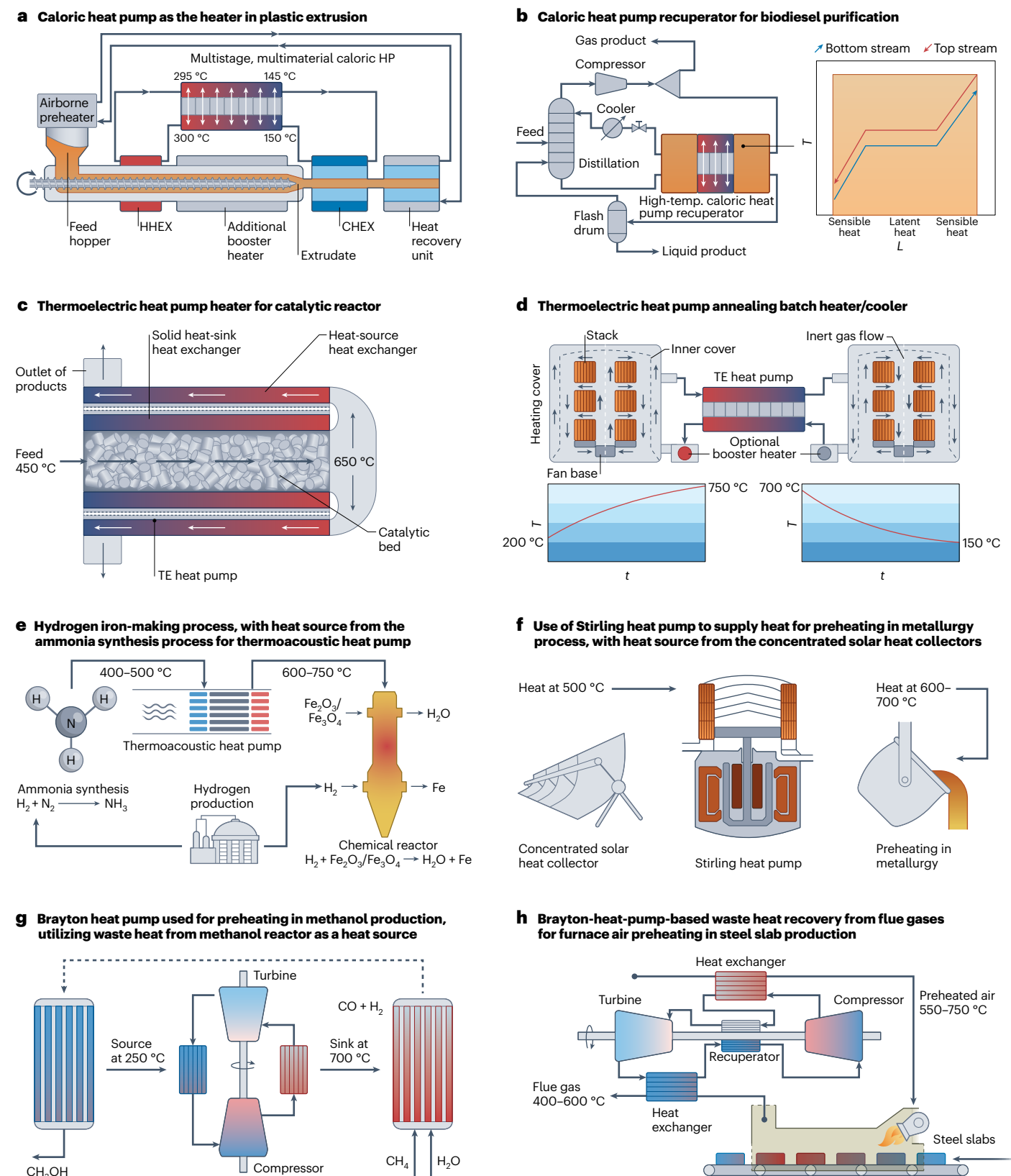


Fig. 6 | Illustration of potential applications for solid-state and gas-cycle high-temperature heat pumps. a–h, Two potential applications each for caloric heat pumps (HP) (a,b) and thermoelectric (TE) heat pumps (c,d), one suggestion each for the application of thermoacoustic (e) and Stirling heat pumps (f) and

two suggestions for the Brayton heat pump (g,h). HHEX, hot heat exchanger; CHEX, cold heat exchanger. Panel b adapted with permission from ref. 101, Elsevier.

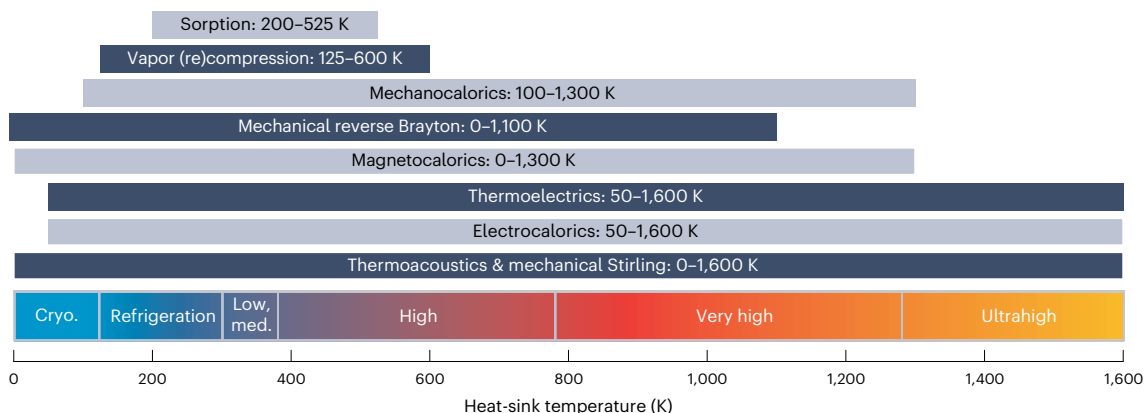


Fig. 7 | Proposed extended classification based on the temperature levels and heat source/sink media used in heat pumps. The new classification covers the whole temperature range of the heat sink from 0 K to 1,600 K, from cryogenic to ultrahigh-temperature heat pumps. The figure shows different combinations of solid, liquid or gas heat sources and heat sinks. The figure also shows the

temperature ranges within which the evaluated technologies can operate. Both the heat-sink and the heat-source media may exist in any phase of matter—solid, liquid or gas—depending on the specific thermophysical properties and boundary conditions of the thermal system.

argon is a sufficiently abundant inert monatomic gas with a high heat capacity ratio, meaning that its temperature increases significantly when compressed, similarly to helium⁸⁸. However, its low-heat-transfer properties and low volumetric heat capacity result in large and robust cycle components. Air and nitrogen are characterized by safety and accumulated turbomachinery design experience. Compared with monatomic gases, greater pressure ratios are required for the same temperature lift. A specific temperature environment can cause nitrogen oxide formation or high-temperature oxidation, mandating the use of effective oxidation-resistant coatings. Carbon dioxide allows for more compact machinery and heat exchangers than other diatomic gases do, but it requires even higher compression ratios and potentially forms corrosive carbonic acid when contaminated with moisture. For temperatures up to 400 °C, ionic fluids can provide enhanced heat transfer from the heat source/sink to the system. Future research must focus on reducing the viscosity and thermal stability at elevated temperatures beyond 400 °C (ref. 89). Liquified metals at temperatures above 500 K (ref. 90) represent a very important future research domain related to working fluids that can serve the heat-source/heat-sink connections to the system. They can be applied in solid-state heat pumps as the primary working fluid or thermal storage medium⁹¹ for design solutions, in which they do not directly interact with the solid refrigerants.

Scaling is the most important problem of caloric technologies. In addition to the required substantial increase in power density per mass of device, new types of field source need to be designed and investigated. Continuous additive manufacturing processes need to be developed for regenerators. Among caloric devices, only specially engineered superconducting magnetocaloric heat pumps²⁹ have the potential for megawatt-scale heating. All other caloric technologies remain confined to outputs up to tens of kilowatts. Although laboratory-scale electrocaloric devices currently represent very limited power, the continuous-flow manufacturing process can boost this technology from the current power of a few experimental devices to at least two-orders-of-magnitude-larger scales while implementing modularity to upgrade the scale. Scaling will further require the introduction of continuous additive manufacturing processes for thermoelectrics and special designs of acoustic generators in thermoacoustics.

Examples for applications

We present a few examples of potential applications of high-temperature solid-state and gas-cycle heat pumps in Fig. 6, with two exemplary applications per evaluated technology. Plastic extrusion systems (Fig. 6a) in the range of a few to tens of kilowatts of heating power are common. In these systems, the cooling process for the extrudate can serve as

the heat source of series-connected electrocaloric heat pumps. The process shown in Fig. 6b involves the high-temperature caloric heat pump used in biodiesel purification, which serves as the recuperator booster. The heat pump is embodied between two heat exchangers (for liquid and for the gas state of the biodiesel) and serves for heat recovery for the latent heat of condensation (top stream) to the latent heat of evaporation (bottom stream). Catalytic heaters (Fig. 6c) can be found in different ranges from 10 W to tens of kilowatts.

The thermoelectric heat pump uses the heat source from the cooling of the products and recovers it for heating the catalytic bed. Another example of the use of high-temperature thermoelectric heat pumps is the annealing process (Fig. 6d), which consists of heating and cooling processes. When two batches are utilized, the process can be considered continuous, that is, it involves switching between heating and cooling. A heat pump can utilize heat from the cooling process as the heat source, upgrade it to the desired temperature level, and use it for heating. The waste heat from the ammonia synthesis process (Fig. 6e), initially at 400–500 °C, is increased to 600–750 °C using a thermoacoustic heat pump.

This high-temperature heat is utilized in chemical reactions in the hydrogen iron-making process. By integrating this system, the waste heat from ammonia synthesis is efficiently recovered. By integrating solar energy into iron-making (Fig. 6f), solar energy is first converted into heat at 500 °C using a concentrated solar heat collector. This heat is then increased to 600–700 °C through a Stirling heat pump and utilized in the metallurgy preheating process. The reversed Brayton heat pump utilizes waste heat from the methanol reactor, where exothermic catalytic conversion of carbon oxides and hydrogen over the Cu/ZnO catalyst is used to produce methanol at temperatures above 210 °C (Fig. 6g). This heat is increased to preheat the methane and high-temperature steam used in the initial phase of methanol production to produce syngas by steam reforming it in the temperature range of 800–900 °C. The integration of a reversed Brayton heat pump into a reheating furnace (Fig. 6h) for steel production enables the utilization of a large amount of available high-grade waste heat from flue gases (heating above 400 °C). This heat is used to preheat combustion air and reduce the consumption of the primary energy source used for steel slab heating.

Need for extended classification

The existing classifications focus on vapour (re)compression heat pumps, neglecting other promising heat-pump technologies and their associated temperature ranges. High-temperature heat pumps necessitate different secondary fluids than those commonly used, such as air–water or water–water, or specific sources, and those used for aerial,

geothermal or solar-assisted heat pumps. These high-temperature heat pumps can utilize different gases, oils, molten salts, liquid metals or direct solid contacts, among others. The advanced heat pumps have diverse applications beyond industrial settings, including potential use in systems or vehicles with internal combustion engines or fuel cells. Our aim is to introduce a classification system based on the heat source or sink: gas, liquid or solid. Additionally, our research indicates that review paper authors often define an ‘ultrahigh temperature’ of approximately 200 °C, raising questions about how to classify heat pumps with, for example, a heat sink temperature of 500 °C or higher. To simplify and ensure the longevity of the classification system, we propose a method that incorporates various heat-pump types (vapour (re)compression, sorption, caloric, thermoelectric and gas cycle) and provides clarity on the type of heat pump being discussed (Fig. 7).

Conclusions

In this Review, we highlight the overlooked potential of high-temperature heat pumps, which, beyond their current market applications, can significantly contribute to industrial process decarbonization. These technologies can help reduce fossil fuel consumption and facilitate the direct conversion of electricity into heat while also utilizing currently untapped waste heat sources. Furthermore, sector coupling through power-to-heat and power-to-heat-to-power systems presents intriguing opportunities for efficiently converting peak electricity into heat and vice versa, especially compared with traditional electric resistive heaters. Decarbonizing heating while simultaneously utilizing waste heat can undoubtedly yield greater benefits than fully decarbonizing the transport sector; thus, we hope that research and innovation in heating decarbonization will increase in the next decades.

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Competing interests

The authors declare no competing interests.

Additional information

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